3.5 Horiz curve

Superelevation

- Mixed traffic or design super elevation.
  - Reduced speed kilni lete h.
  - e formula
    - in ms^{-1}
    - in kmph

4) Superelevation for mixed traffic condition or design superelevation

$1v \longrightarrow 0.75v$

e at flat $= \frac{v^2}{gR} \longrightarrow 0.75v$

$e = \frac{(0.75v)^2}{gR}$

$e = \frac{V^2 \longrightarrow kmph}{225R}$

lateral = 0.15 lete h. in general case ⓑ me.

ki value generally kyalete h!

In case of mixed traffic condition the speed of vehicle remains 75% of its original speed. ie $\frac{3}{4}v$ or $.75v$

Design of superelevation steps

Steps

- Step ① - mixed traffic
  - e
    - general traffic
- Step ②
  - Condns
    - $e \leq e_{max}$

Step ① - (mixed traffic) $e = \frac{v^2}{225R}$

Step ② :- calculate general $e = \frac{v^2}{127R}$

Step ③ :-

(i) If $e < e_{max}$

3.1) Design of superelevation

Step 4 v calculation

- restricted speed kitni denge.

Step 4

$$e_{max} + f_{lat} = \frac{V^2}{127R}$$

$$e_{step\,②} + 0.15 = \frac{V^2}{127R}$$

$\sqrt{127R(e+0.15)} = V_{restricted\ speed}$

Determining radius

- ruling min radius
  - kyah, kaise derive karte hai
  - kya use hai
  - formula

Determination of radius

① Ruling minimum Radius

It can be derived by maximum superelevation & maximum coeff of lateral friction. With design speed.

Ideally the radius of curve should be more than ruling radius

$$e_{max} + f_{lat\ max} = \frac{V^2}{127 R_{Ruling}}$$

IRC 0.15 IRC

Determin of radius

② abs min radius

- kaise derive karteh.

- formula.

V is design speed of vehicle

① absolute minimum radius. (R_{abs})

Absolute minimum radius is obtained @ maximum superelevation & maximum coefficient of friction with minimum speed.

f_{max} + f_{latera} : $\frac{V_{min}^2}{127 R_{abs}}$

IRC 0.15

IRC

Horizontal curve extra widening design

- extra widening kyaheh

- psychological kyaheh

Horizontal Curve [extra widening]

. Extra widening on horizontal curve refers to additional width of carriageway ie required on curve from psychological aspect & mechanical aspect.

Psychological widening

. There is a tendency for the drivers to drive close to the outer edges of pavement

3.9 Extra widening

- Psychological widening
- formula.

hence some clearance or space is to be provided for crossing and overturning opens

$$Wps = \frac{V}{9.5\sqrt{R}}$$

R is radius of circular curve in m

V is velocity in kmph

W is obtaind in m.

- Off tracking
- formula
- kya hoti hai
- kiske liye defined hai

Off tracking

formula

off tracking $= \left(\frac{l^2}{2R}\right)$

when a vehicle negotiates horizontal curve the rear wheel follows a path of shorter radius than front wheel, this phenomena is called off tracking.

- off tracking is defined for single lane

Extra widening
- Mechanical widening
  - formula
    - nkyas
    - lkyas
- Combined formula
- Sharp curve pe kahan dete h?
- Normal curve pe kaha dete h?

③ Mechanical widening

W_m : no of lanes × ff tracking

$$: \frac{n l^2}{2R}$$

W_m : $\frac{n \text{ nos}}{2R} \frac{l^2}{\text{m}}$ m

n

l is length of rigid wheelbase

n is no of lane of largest vehicle ($l = 6.1$ m)

④ Total extra widening

$$W_{extra} = W_m + W_{ps}$$

$$W_{extra} : \frac{n l^2}{2R} + \frac{V}{9.5\sqrt{R}}$$

V → kmph

W_{extra} → m

Note : on sharp curves extra widening is provided fully on inner side.

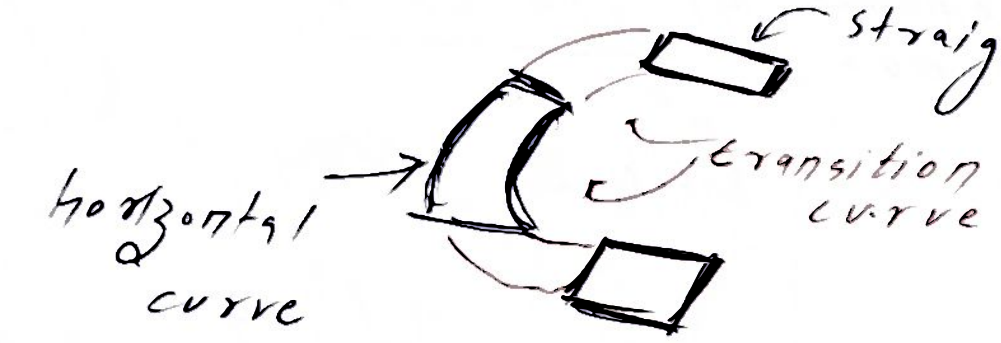
on normal road extra widening is given equally on both sides at uniform rate.

- Horizontal curve design
  - alternate name
    - transition curve milb kya h?
    - Transn curve kahan de te h?
  - Diagram

Horizontal curve or Transition curve design.

Transition curve ka matlab highway engg me horizontal curve ko bola jata h. ↕ a little contradictory

Transition curve is a curve which is introduced between circular curve and straight line.



- Types
  - Spiral
    - alternate names
    - Speciality
    - length Radius relation proportionality
    - rate of chg of acc kaisa h?

Types / Shaps

1. Spiral or clothoid or glovers Spiral

- Ideal shape for transn curve
- LR $L \propto R^{-1}$
- rate of change of acceleration is uniform.

- Curve
  - types
    - Bernoulli's lemniscate
      - Used where
      - Radius length relation.
      - Speciality
    - Cubic parabola
      - alternate name
      - used where
      - formula
    - Cubic spiral
      - speciality

(ii) Bernoulli's lemniscate

- Mostly used in modern roads where deflection angle in curve is large
- Radius of curve decreases more rapidly with length.
- It is an autogenous curve.

(iii) Cubic parabola or trouded easement curve

- used for valley curve
- $y = L^3 / 6RL_s$ L_s is length of SSD

(iv) Cubic spiral

- simple to set curve

4.1) kyo transition curve defeth?

- Grad in the
- steerg
- c

Class 4 7 Aug 24

Objecl of pooriding transilion curve

. to introduce gradually centrifuga l force between the tangent points and beginning of circular curve to avoid jerk.

. to enalle the driver to turn the steering gradually for comfort and security

. it introduces the superelevation and extra widening on curve gradually

ength of transcurve

Limiting length of transition curve

. . . .

- Rate of Δ of centr acc
  - formula
    - Cinms'
      - rang e of c

rate of change of radial or centrifugal acceleration

$L_s = \frac{V^3}{CR}$

V → velocity in ms⁻¹

R → radius of curve m

C → rate of change of centrifugal acceleration

length of transn curve

- c formula

$$C = \frac{80}{75 + V}$$

C → in m/s^3 ; V → in kmph

- c range

$-0.5 \leq c < 0.5$ ← range of c.

② Rate of Δf

Rate of change of superelevation.

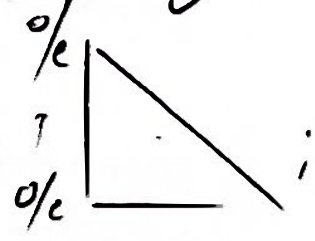
(i) rot abt inner edg

(i) if pavement is rotated about inner edge (rotated outer edge)

- formula

$$L_s = e N \left[W + W_e \right]$$

e → superelevn; N → rate; W → width; W_e → extra widening



(ii) rot about centre line

(ii) When pavement is rotated about centre line (rotated inner & outer edge)

- formula

$$L_s = e N \frac{\left[W + W_e \right]}{2}$$



e is superelevation

We is extra widening

length = superelevn × rate of chg of selevn × [pvmt width + extra widening / 2]

1 : N or 1 in N : rate of change of superelevation

A road 10m wide is deflected through an angle of 65° with centre line radius 350m a transition curve is used at each end of circular curve such that the gain in rated change of radial acceleration is 0.4 ms^{-3}. When the speed is 60 kmph find length of transition curve and shift

Ans given data

$V = 60\ kmph = \frac{60 \times 1000\ m}{60 \times 60\ sec} = 16.67\ ms^{-1}$

$\theta = 65°$

$C = 0.4\ m/sec^3$

$R = 350\ m$

$L_s = \frac{Velocity^3}{Change\ in\ rate} \times Radius$

$= \frac{(16.6)^3}{0.4 \times 350} = 33.07\ m$

$S = \frac{Length\ of\ Xn\ curve^2}{24 \times Radius} = \frac{33.07}{24 \times 350}$

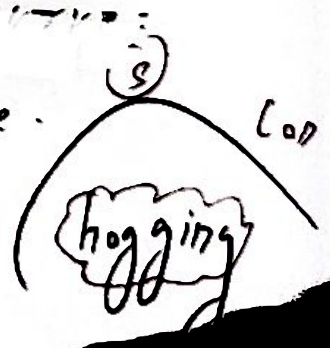
- Vertical curve
  - Provided where?
  - Summit curve
    - alternate names A, h, C

Vertical Curve

- Provided in the elevation @ change of gradient (level or vertical level)

1) Summit curve

- is called (check) ascending curve
- also called hogging curve.
- Convex curve



4.3 Length of Xs curve

N as per IRC

N value in plain & rolling terrain — N 150 in plain and rolling terrain

N value in hill & mountainous — N 60 in hill & mountainous terrain

W is width of pavement.

(3) empirical formula as per IRC

(3) Empirical formula as per IRC

(i) plain & rolling terrain

$L_s = \frac{2.7V^2}{R}$ or $\frac{2.7 (\text{Velocity in km})^2}{\text{Radius of Xs curve}}$

- plain & rolling terrain formula

- mountainous or steep terrain

(ii) Mountainous or steep terrain

$L_s = \frac{v^2}{R}$ or $\frac{\text{velocity}^2}{\text{radius}}$

- Teeno me se criteria me se kaunsa large finally

among all 3 criteria (maximum) length is considered

① ② ③ } ↑ max

Shift of curve

- formula

- what is L_s

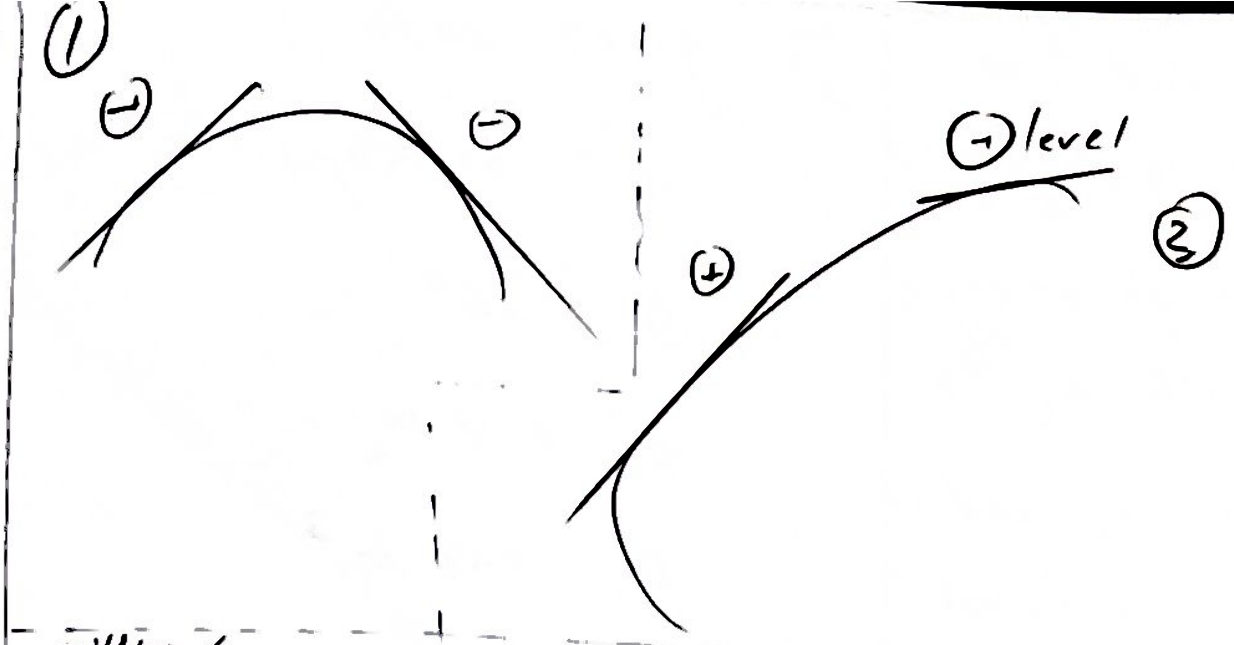
Shift of curve

$S = \frac{L_s^2}{24R}$ or $\frac{(\text{Length of trans curve})^2}{24 \times \text{Radius}}$

4.5 • Vertical curve

S. ① Summit curve

- Diagrams



- kab kab detch
  - + -
  - + mild
  - + 0
  - - steeper
- length of summit curve (Design)

Summit curve is provided when

① +ve grade [ascending grade] meets -ve grade [descending grade]

② +ve grade meets another mild +ve grade

③ When +ve grade meets level sketch

④ When -ve grade meets steeper -ve grade

length of summit curve design.

1. Design criteria based on SSD.

length of summit curve design

- Based on ssd
  - assume $L \gg$
  - formula
    - H kyak
    - h kyak
    - h_1 kyak
  - formula when std values are taken
- How to calculate geometric difference in grades or deviation angle.

(a) assume $L \geq SSD$

$$L = \frac{NS^2}{\left(\sqrt{2H} + \sqrt{2h}\right)^2}$$

$$= \frac{\text{deviation angle} \times SSD^2}{\left[\sqrt{2 \times \text{driver's eye ht}} + \sqrt{\text{obstruct}^n \text{ ht}}\right]^2}$$

$$L = \frac{NS^2}{\left[\sqrt{2 \times 1.2} + \sqrt{2 \times 0.15}\right]^2}$$

$$L = \frac{NS^2}{4.4}$$

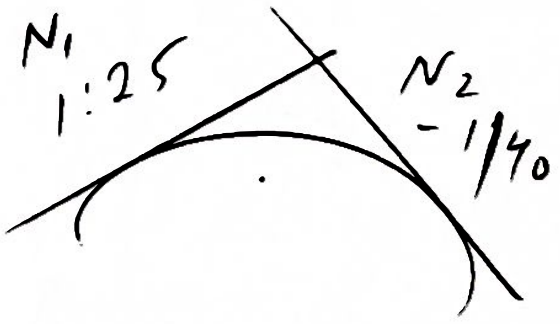
N is deviation angle or algebraic difference in grades



$$N = |+3 - (-5)|$$

$$= 8\%$$

$$= 8/100$$



$$N = |N_1 - N_2|$$

$$= \left|\frac{1}{25} - \left(\frac{-1}{40}\right)\right|$$

$$= \left|\frac{1}{25} + \frac{1}{40}\right|$$

$$= 0.065$$

length of summit curve

- OSD
  - assume L > OSD
    - formula
    - formula for std value
    - Check
      - Pass
      - fail
    - formula
    - formula using std values

① Design criteria based on OSD (also valid for ISD)

(a) assume $L > OSD$ or ISD

$$L = \frac{NS^2}{\left[\sqrt{2H} + \sqrt{2H}\right]^2}$$

$$L = \frac{NS^2}{9.6}$$ Yahan S ko OSD ya ISD lenge.

Check

— Pass OK ✓ → Provide the calculated L

— fail Not OK (b) $L \not> OSD$ or ISD
$L < ISD/OSD$

$$L = 2S - \frac{\left[\sqrt{2H} + \sqrt{2H}\right]^2}{H}$$

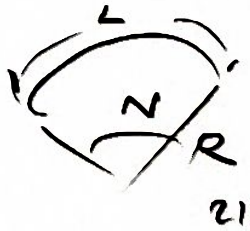
$$= 2S - \frac{9.6}{N}$$

Minradius of parabolic summit curve formula.

Note minimum radius of parabolic summit curve

$$R = \frac{L}{N}$$

angle = $\frac{arc}{Radius}$



U-9 Summit curve

- Best shape?
- eqn of sq pbla

Best shape for summit curve is square parabola.

Equation of square parabola $y = \frac{Nx^2}{2L}$

Valley curve

- alternate names
- Best shape for valley curve all names.
- use
- Design criteria
  - HLD
  - M comft
  - Cross drng ctrl
  - Gen aes

Vertical curve (Valley curve)

. Sag curve / Concave crve

. Best shape for valley curve is cubic parabola / easement curve / froudes transition curve

. use: purane wale me jahan +ve tha woh -ve aur -ve tha woh +ve ho jayega / summit ke opposite

• Design criteria

① . Vehicle headlight distance

② . Motorist comfort.

③ Cross drainage control @ lowest pt of valley

④ General aesthetics

Design of valley curve

- Comfort crit
  - kyd discomfort hotagh?
  - formula
    - C valu e?
    - Std formula wit C included
- ② HSD
  - assume L > HSD
    - L formula
      - B kyq h2?
      - Std formula

Design.

① Comfort criteria

. In case of valley curve the pressure on engine or spring are more due to centrifugal force and gravity or weight of vehicle all acting downward. Hence it causes discomfort to driver.

$$L = 2 \times \sqrt{\frac{N v^3}{C}}$$

put $C = 0.61 \text{ m/s}^3$ $v = 0.278\,V$ ← we get

$$L = 0.38 \sqrt{N V^3} \quad \text{kmph}$$

② Design based on headlight sight distance (also valid for SSD)

ⓐ assume $L > HSD$

formula $L = \dfrac{N S^2}{[2h_1 + 2 S \tan\beta]}$

h_1 = head light beam L = .75

$\tan 1^\circ = 0.017$

$$L = \frac{N S^2}{2 \times 0.75 + 2 \times 0.017} = \frac{N S^2}{1.5 + 0.034 S}$$

∝ std formula

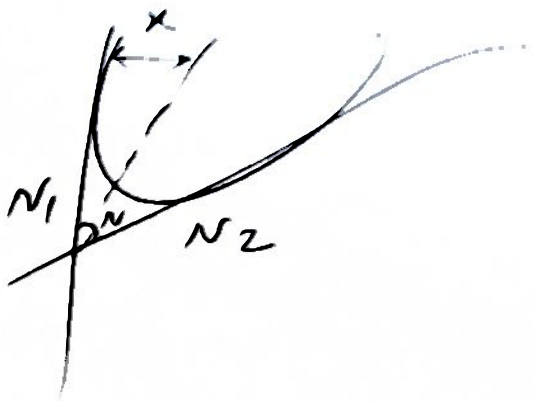
214

Centrifugal ratio

N_1 is gradient of initial tangent

N is deviation angle; x is location of deepest or deepmost point from initial tangent.

L is length of valley curve



location of deepmostpt = $\frac{\text{gradient of initial tangent} \times \text{length of valley curve}}{\text{Deviation angle}}$

- alternate name — Impact factor or centrifugal ratio
- kya h 2.
- formula & alternate formula — I = P/W or I = $1.6NV^2/L$
- Impact factor kitna zyada rakhte hain — Impact factor cannot be more than 17% as per IRC

Vertical curve design

- HSD
  - assume L > HSD
    - Check pass
    - Check fail
      - formula
      - std formula

Check

- if pass OK L > HSD
  - ⓐ L ✓
- if fail Not OK
  - ⓑ

ⓑ $L \neq HSD$ or $L < HSD$

$$L = 2S - \frac{(2h_1 + 2S\tan\beta)}{N}$$

$$L = 2S - \frac{1.5 + 0.034S}{N}$$

- Why is OSD not a problem in Vertical curve.

Note OSD: overtaking sight distance is not a problem for valley curve because in daytime visibility is clear & during nighttime headlights distance can be seen from a considerable distance.

- Position of deepmostpt in valley curve formula.

. The location of deepest or deep most point in valley curve $x = \frac{N_1 L}{N}$

5

Curvere sistance

- kya h?

- kin vehicles me hota h

- Hills pe kaisa hota h

- formula
  - T kyah
  - θ kyah

- Tractive force definition

Setback, Curve resistance & Grade compensation

Class 5

8 Aug 2024

Curve resistance

Curve resistance is the less in tractive force / traction effort due to turning of vehicle on a horizontal curve.

- Curve resistance develops only in vehicles having rear driving wheels
- Curve resistance is acute in hillroads

$\Delta T = T - T\cos\theta$

T is actual tractive force

θ is turning angle

- Tractive force: force needed to overcum the resistance caused by friction when two bodies slide on eachother

Setback / Setback distance M

- Setback
  - alternate name
  - Defn
  - approximate formula
  - factors affecting
    - 1
    - 2
    - 3
  - Rational formula

- Clearance

. It is the distance required from the centreline of horizontal curve to an obstruction on the inner side of curve to provide adequate sight distance @ a horizontal curve.

$M = L^2/8R$ — approximate

or $\frac{\text{Length of curve for SD}}{8 \times \text{Radius}}$

Setback distance $M = \frac{S^2}{8R}$

or $\frac{\text{sight distance desired}^2}{8 \times \text{Radius of curve}}$

factors affecting setback distance

1. Sight distance
2. length of curve
3. Radius of horizontal curve.

Rational formula

6.3 Setback

Rational formula

- ① L ≥ S
  - Single lane
  - multi lane
    - d
      - key
      - diagram
- L < S
  - for multi for α

① $L \geq S$

↓ single lane

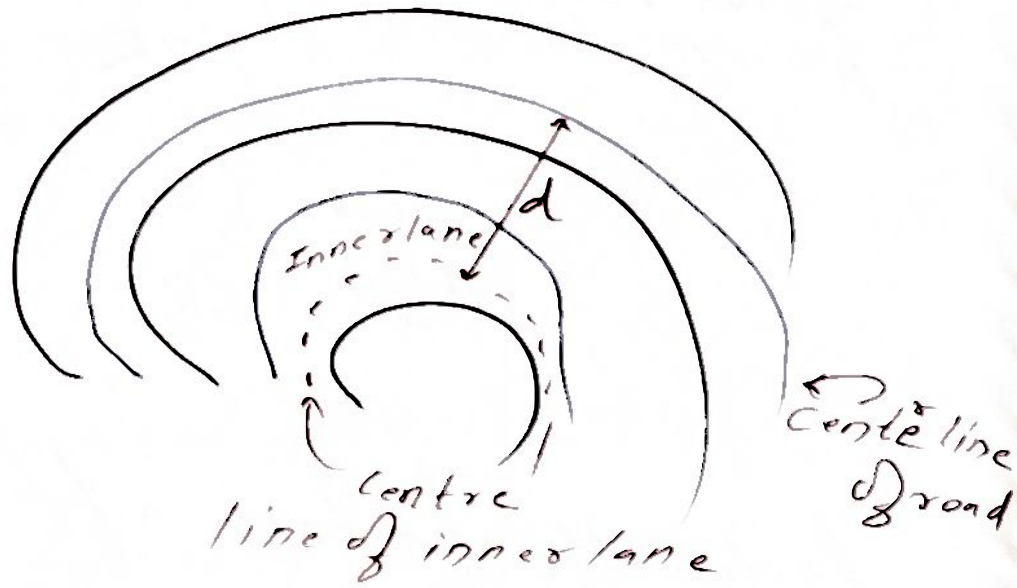
$$M = m = \left(R - R\cos\frac{\alpha}{2}\right)$$



if multilane

$$M = m = R - (R-d)\cos\frac{\alpha}{2}$$

d is distance between centre line of road & centre line of inner lane.



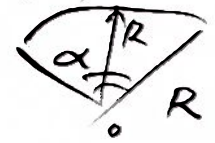
② When $L < S$ (not imp)

Single lane $M = \left(R - R\cos\frac{\alpha}{2}\right) + \left(\frac{S-L}{2}\right)\sin\left(\frac{\alpha}{2}\right)$

multilane $M = R - (R-d)\cos\frac{\alpha}{2}$

$$+\left[\frac{S-L}{2}\right]\sin\left(\frac{\alpha}{2}\right)$$

$\frac{\alpha}{2} = \frac{180}{2\pi R}\, S$ ← single lane



Setback
- Rational formula for multilane
LHS (not imp)

$$\frac{\alpha}{2} = \frac{180 S}{2\pi (R-d)} \leftarrow \text{for multilane}$$

S is sight distance
R is Radius of curve fm centre line of road.
m is clearance or setback.

Grade compensation
- Defn, kab dete hai

Grade compensation

The reduction given to the gradient when a gradient comes along with horizontal curve as per IRC grade compensation should be provided

- kab naidenge

Grade compensation should not be given to the gradient less than or flatter than 4%. G = 4%.

- formula.

formula G: gradient or grade or longitudinal slope (%)

GC: Grade compensation.

$$GC_{(\%)} = \left[\frac{30 + R}{R}\right] \%$$

$$\left[\frac{75}{R}\right]$$

R is rad of curve in m

5.1) Grade compensation

- CG formula

CG: compensated grade

$= (G - GC)$ %.

Q Gradient on a highway is 1 in 15. Radius of curve is 150 metre. ~~after~~ find out grade compensation & compensated grade.

Ans: G = 1 in 15

R = 150 M.

$G = \frac{1}{15} \times 100 = 6.67\%$

[check it is more than 4% hence GC & CG are provided]

$GC = \frac{30 + R}{R} = \frac{180}{150} = 1.2\%$

$= \frac{75}{150} = 0.5\%$

GC = 0.5%.

CG = G - GC = 6.67 - 0.5 = 6.17.

Creeperlane:

- kya hota h?

Creeperlane: When the uphill climb is extremely long, it may be desirable to introduce an additional lane so as to allow slow ascending vehicles to be removed from main stream so that fast moving vehicles are not affected.

frontage road defn.

Frontage road: it is provided to give access to the properties along an important highway with controlled access as expressway or freeway.

Minimum width of shoulder kya hogi

Minimum width of shoulder recommended by IRC : 2.5m.

Guard rail kya hota h kab dete h?

Guard rail is provided when embankment height is more than 3 m.

Q1 Worksheet 1 ka qn 1.
Speed of overtaking & overtaken vehicle are 100 & 50kmph resp....
avg acc 0.92 m/s^2
find OSD
desirable length of ot zone

Ans 1: OSD: $d_1 + d_2 + d_3$

Numericals: $0.278 V_b \times 2 + 2s + b + 0.278 V_c T$.

V_b : 50 kmph

$d_1 = 0.278 \times 50 \times 2 = 27.8$ m.

$T = \sqrt{\frac{4s}{a}}$ $s = 0.7 V_b + 2$

$s = 0.7 \times 50 + 6.1 = 41.1$ m

$= \sqrt{\frac{4 \times 41.1}{0.92}} = 13.36$ second

$2s = 2 \times 41.1 = 82.2$ m.

$b = .278 V_b T = 0.278 \times 50 \times 13.36$

$b = 185.7$ m

5-7

$d_2 = 2sV + 6$

$= 82.2 + 185.7$

$d_2 = 267.8\ m.$

$d_3 = 0.278\ V_c\ T$

→ Va le lo.

$d_3 = 0.278 \times 100 \times 13.36 = 371.4\ m.$

$d = d_1 + d_2 + d_3 = 185.7 + 267.8 + 371.4$

$= 666\ m.$

Desirable length $\geq$ OSD $\overset{5x}{=} 666$

$= 3330\ m$

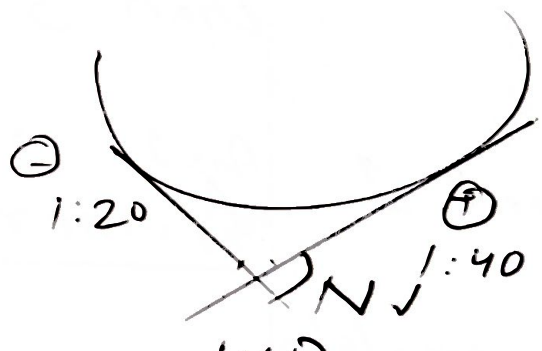
Ans

Q. Worksheet 1 ka Qn 5.

A falling gradient of 1in20 meets rising gradient of 1in 40 on ndr in plain country $V_s = 120$. find length of valley curve which should provide safe driving distance.

(i) $L \geq HSD$

$L = \frac{NS^2}{1.5 + 0.34S}$



gn me s naidiya
h toh apan
s = 120 m leggc.

$N_1 = 1 \text{ in } 20 = \frac{-1}{20} = -0.05$

$N_2 = +\frac{1}{40} = +0.025$

N is difference or deflection L

$= |N_1 - N_2| = |-0.05 - 0.025|$

$= +|-0.075| = 0.075$

$$L : \frac{NS^2}{1.5 + 0.034S} = \frac{0.075 \times 120^2}{1.5 + 0.034 \times 120}$$

$$= 193.5\ m.$$

Check: $0L > S$

$193.5 > 120$ Pass.

Ans is 193.5 Mk.

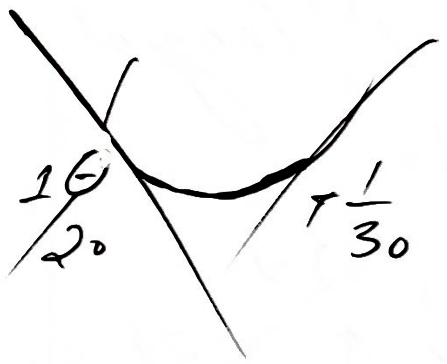
for the same question or above qn if $S = 220\ m$ then what will be length of valley curve.

---

8 A valley curve is found or formed by a descending gradient $\frac{1}{20}$ meet with an ascending gradient $\frac{1}{30}$. The design speed of vehicle is 80 kmph the allowable rate of change of centrifugal acceleration is taken as $0.61\ ms^{-3}$. find out length of valley curve for comfort condition.

Ans:

$$N = |N_1 - N_2|$$



$$N_1 = \frac{-1}{20} = -0.05$$

$$N_2 = +\frac{1}{30} = +0.033$$

$$N = |-0.05 - 0.033| = 0.083$$

$C = 0.61\ ms^{-3}$

$V = 80$ kmph.

$$L = 0.38\sqrt{NV^3}$$

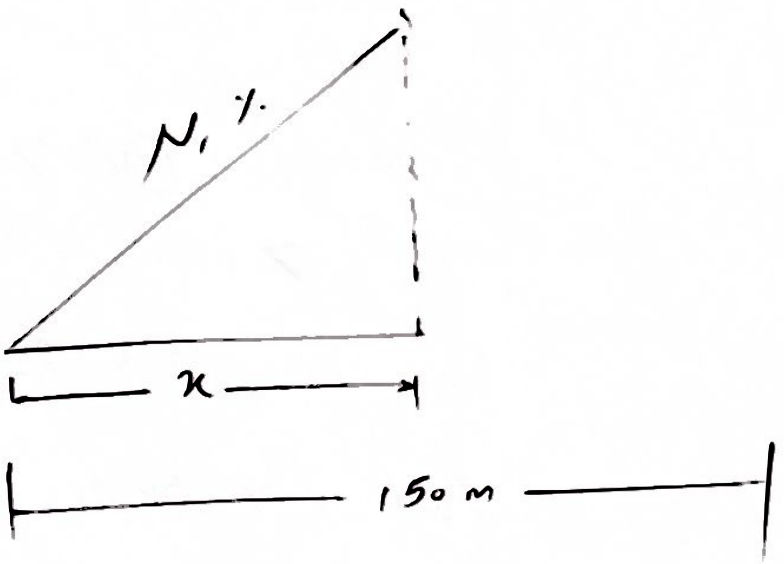
$$= 0.38\sqrt{0.083 \times 80^3}$$

$$= 78.33\ m \quad \text{Ans.}$$

Class 6, 9 Aug 2024

E.1) A rising gradient of a parabolic curve is 4% & falling grade is 2%. Then the highest point on the summit curve is at a distance measured horizontally from 1st tangent will be.

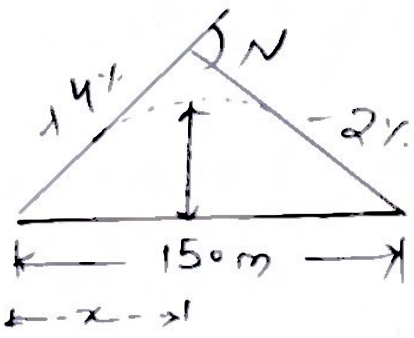
Ans: Parabolic, summit curve



$N = |N_1 - N_2|$

$= |4 - (-2)|$

$= 6\%$ or 0.06



$\frac{N\%}{L} = \frac{N_1\%}{x} \Rightarrow x = \frac{N_1\% \cdot L}{N}$

$= \frac{(4/100) \times 150}{(6/100)} = 100\ m$

Gate ka qn. Very imp.

Q2) A rising gradient 1 in 50 meets a falling gradient 1 in 100 the length of vertical curve if rate of changed gradient is 1% per 100m length is.

$N_1 = 1:50$

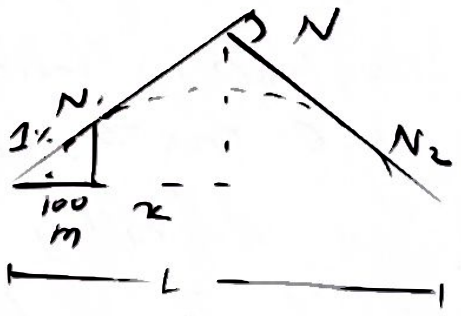
$= 2\%$

$N_2 = -1:100$

1% per 100m

$L = ?$

$N = \left| \frac{1}{50} - \left(-\frac{1}{100}\right) \right| = 0.03 = 3\%$



$$\frac{L}{N} : \frac{x}{N_1} : \frac{100m}{1\%}$$

$$\frac{L}{0.03} : \frac{100}{0.01} : L : 300m$$

alternate method

1% percent $\rightarrow$ distance 100m

3% per distance, $\frac{3\%}{1\%} \times 100$

$= 300m$

Calculate SSD for V = 50 kmph for

(1) 2 way traffic for 2 lane road

(2) 2 way traffic in single lane road

Note: Consider values as per IRC

Ans: $SSD = 0.278 V_0 t_R + \frac{V^2}{254(f \pm n\%)}$

$f = 0.35$

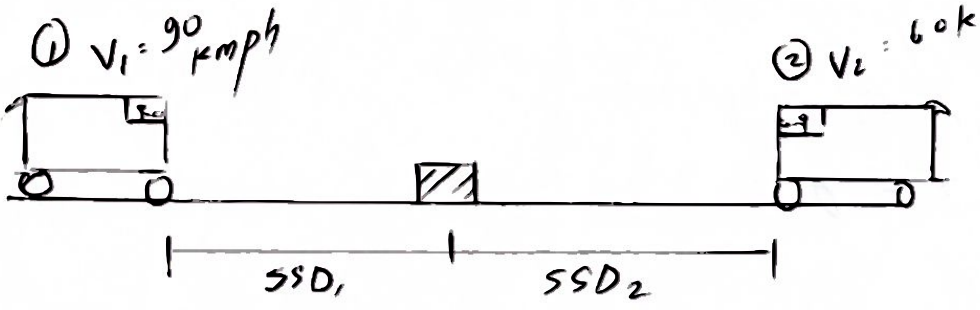
$t_R = 2.5$

$= 0.278 \times 50 \times 2.5 + \frac{50^2}{254 \times (0.35)}$

$= 61.4 \approx 62.8 m$

For (1) SSD: 62.8m for (2) SSD: 2SSD = 125m

What is the minimum sight distance to avoid head on collision of two cars approaching @ 90 kmph and 60 kmph speed use reaction time 2.5 second brake $f = 0.35$ in either case $f = 0.7$ coeff of longitudnal friction



$SSD = V t_R + \frac{V^2}{2g(f \pm N\%)}$

$SSD_1 = 0.278 V t_R + \frac{V^2}{2g(f \pm N\%)}$

90 → 25 → 0